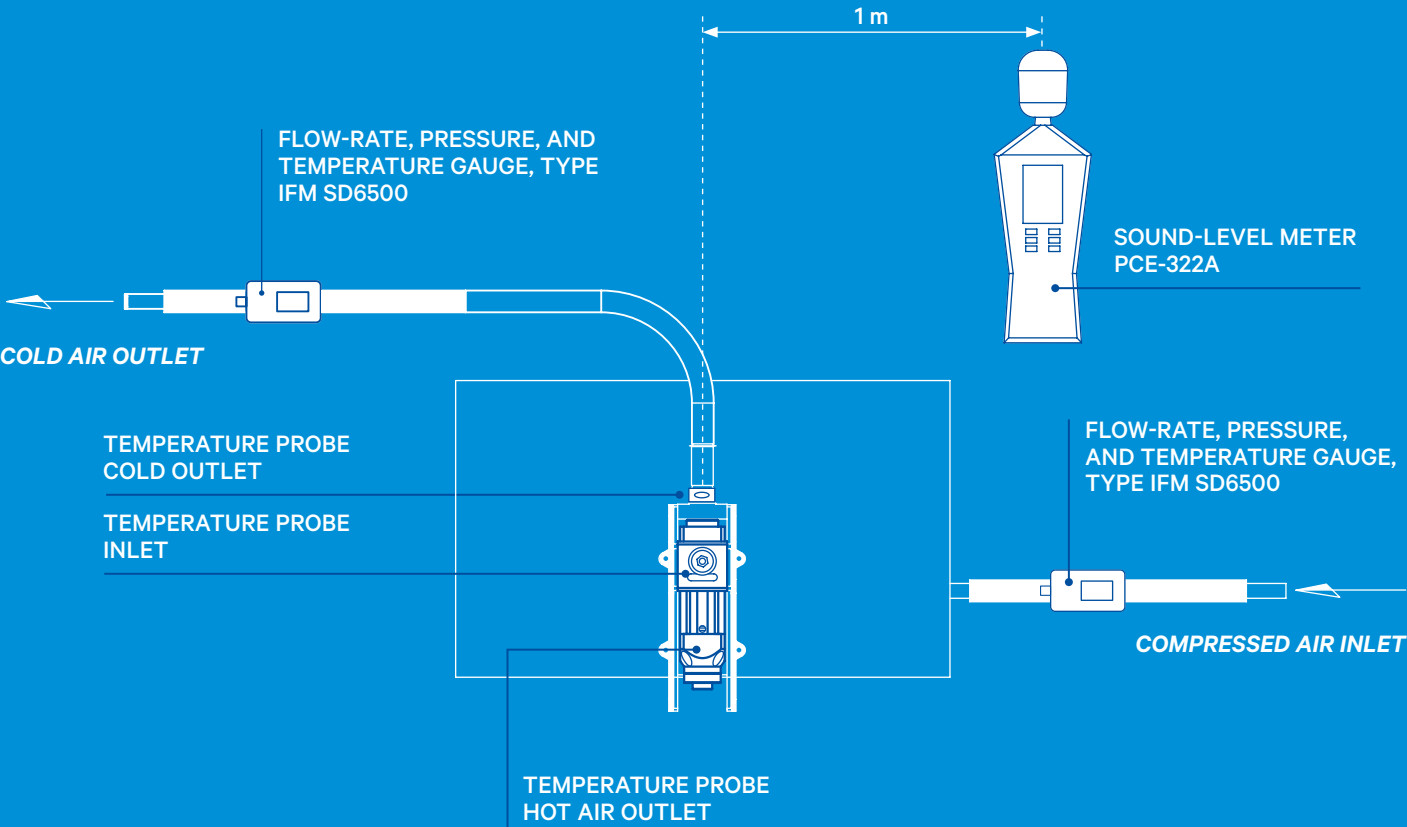
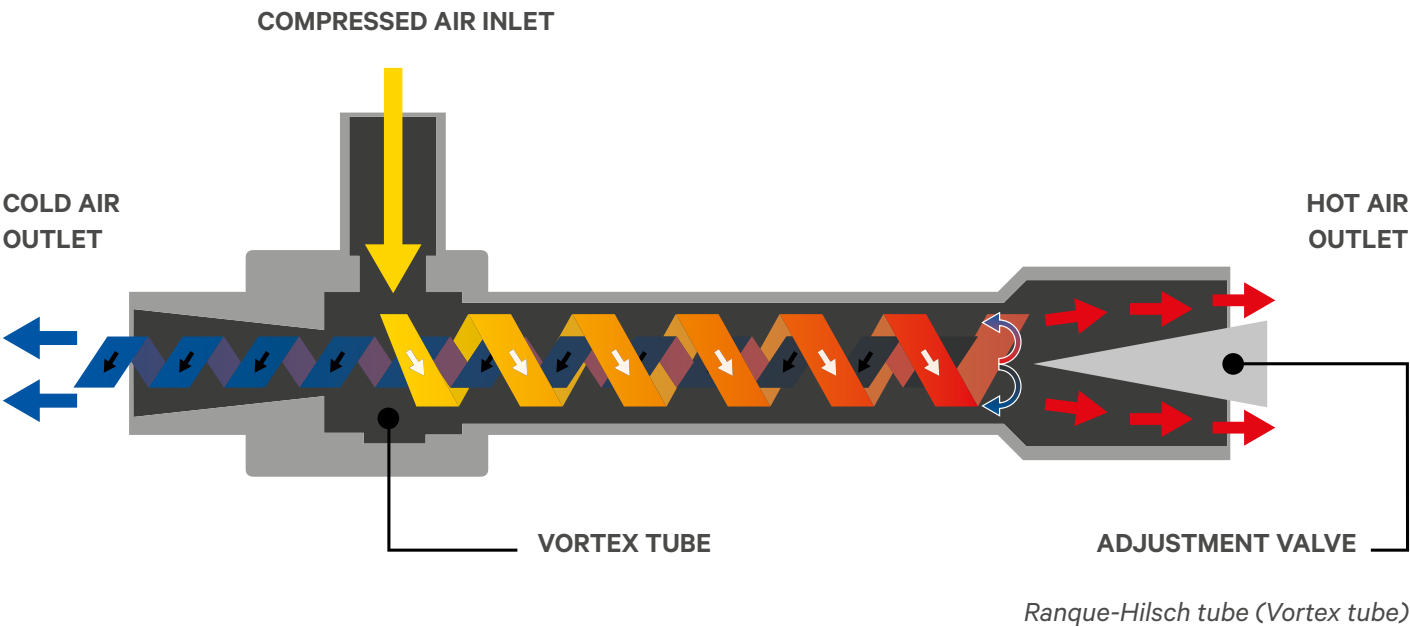


Description of the set-up and instrumentation used for our tests on pneumatic coolers.



The **VR / VRX / VR U-G SERIES** coolers are state-of-the-art solutions for compressed-air cooling based on the principle of the Vortex Tube. The excellent performances of flow-rate and ΔT generated, the design, the fastenings that make them extremely versatile to mount, and the possibility to combine them in a patented system with the air amplifiers (to use the hot air flow), offer customers an innovative, effective, and inexpensive solution to cool down metal and plastic parts, electric and electronic control cabinets, and mechanical applications. All this with a simple connection to the compressed-air line.

- ΔT up to -40°C for the cold flow and $+60^{\circ}\text{C}$ for the hot flow, in comparison to the temperature of air at inlet
- Easy to install, thanks to flanges and magnetic supports
- Patented system of hot air's recovery to actuate an amplifier/conveyor
- Made of corrosion-resistant materials
- No moving part, so not subject to wear and tear
- No electricity or chemical substances required
- They do not cause either sparkles or interferences
- Instant operation
- Reliable and maintenance-free



DESCRIPTION OF VORTEX TUBES

The Ranque-Hilsch tube, in the industrial sector better known as “Vortex tube”, is a device that splits a compressed-air flow in 2 separate streams: one of cold air, and one of hot air.

The core of the system is the vortex chamber, which is connected to 2 opposed tubes, one of which features a valve. When the compressed air is injected tangentially in the chamber, this causes the rotary movement of air towards one of the exits. This vortex moves rotating at high speed and brushing against the inner side of the tube, increasing in temperature; the valve placed at the hot air outlet enables some of it to be exhausted. The remaining part goes back, creating a low pressure vortex moving towards the other exit and giving away heat to the first vortex. So, this flow is much colder.

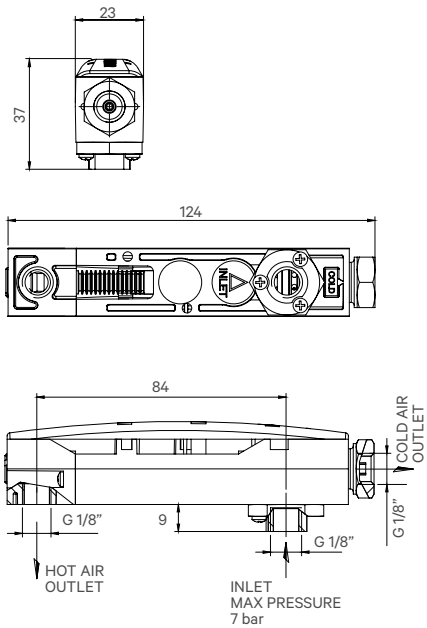
The ΔT generated is inversely proportional to the volume of the flow. The differences in temperature are considerable and can reach -40°C for the cold flow and 60°C for the hot flow.

In the industrial field the Vortex tubes have been employed for a long time and have found a variety of applications in which they offer a major added value. They have great cooling performances, are very easy to install and have instant operation, have no moving part and therefore are maintenance-free. Plus, they do not require electric power, so they are suitable for dangerous environments and humid areas.

If the application enables their use, they are price-worthier than electric coolers. Our coolers SERIES VR and VRX, beside the excellent performances in comparison to the other products in the market, were designed to be easily customised according to customers’ demands.

SERIES VR-100

MODULAR PNEUMATIC COOLERS



GENERAL FEATURES - VR-100

Materials	Body and cover: Nylon 6.6
	Air connections and nozzles: brass
Air inlet port	G-1/8" F
Outlet port (cold flow)	G-1/8" F
Exhaust port (hot flow)	G-1/8" F
Recommended hose	Ø-8x1
Air supply pressure	3 ÷ 7 bar
Cooling power*	120 W - 100 Kcal/h - 400 BTUH
Optional magnetic support	KACM-VR100

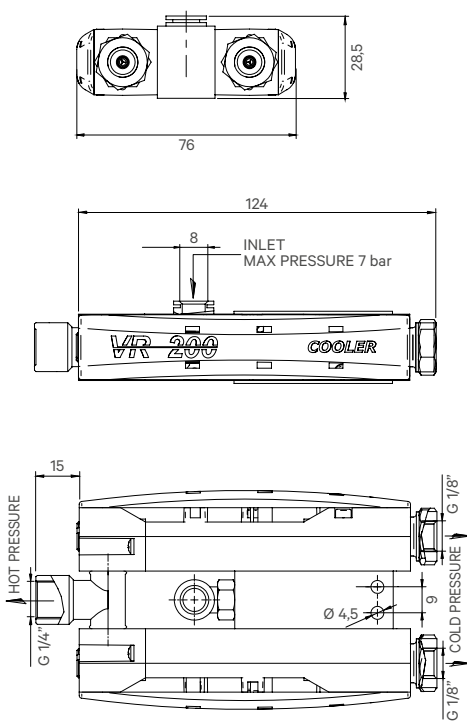
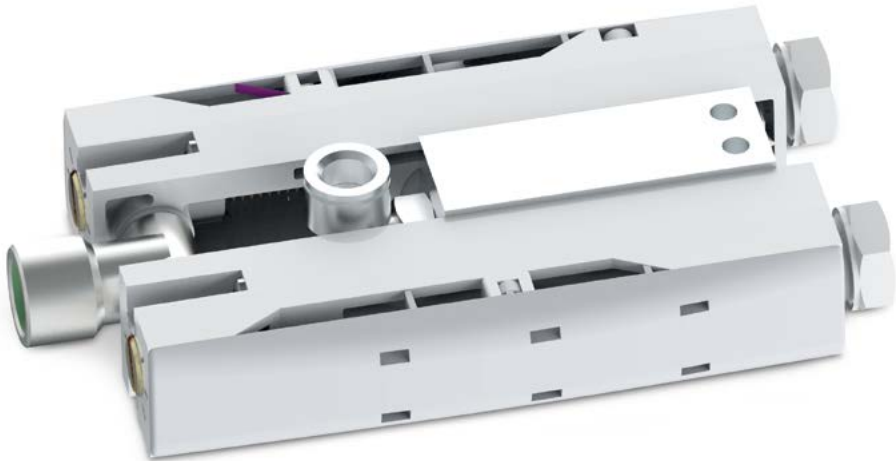
*with inlet pressure 7 Bar and inlet temperature 20°C

PERFORMANCES AND CONSUMPTION TABLE (with air temperature at inlet 20°C)

Pressure bar	Outlet temperature cold flow °C	Consumption NL/min
1	-1.5	32
2	-8	53
3	-15	74
4	-21,5	94
5	-24,5	115
6	-26,5	135
7	-28	154

SERIES VR-200

MODULAR PNEUMATIC COOLERS



GENERAL FEATURES - VR-200

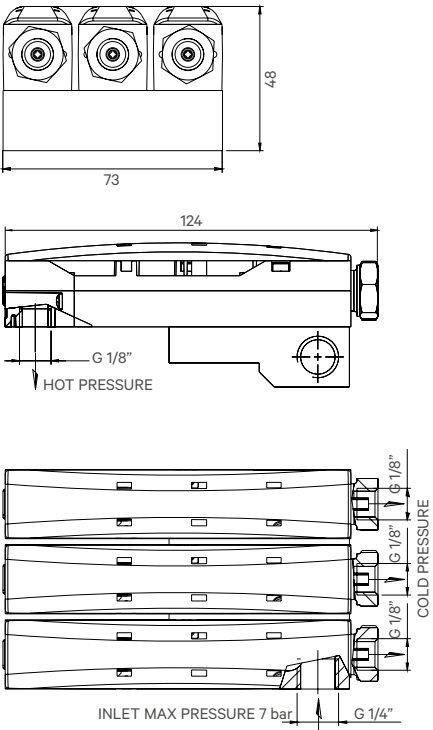
Materials	Body and cover: Nylon 6.6
	Air connections and nozzles: brass
Air inlet port	Push-in fitting Ø-8x6
Outlet port (cold flow)	2 x G-1/8" F
Exhaust port (hot flow)	2 x G-1/8" F
Recommended hose	Ø-8x1
Air supply pressure	3 ÷ 7 bar
Cooling power*	240 W - 200 Kcal/h - 800 BTUH
Optional magnetic support	KACM-VR200

*with inlet pressure 7 Bar and inlet temperature 20°C

PERFORMANCES AND CONSUMPTION TABLE (with air temperature at inlet 20°C)

Pressure bar	Outlet temperature cold flow °C	Consumption NL/min
1	-1,5	64
2	-8	106
3	-15	148
4	-21,5	196
5	-24,5	230
6	-26,5	270
7	-28	308

SERIES VR-300T • 3 OUTLETS
MODULAR PNEUMATIC COOLERS



GENERAL FEATURES - VR-300T

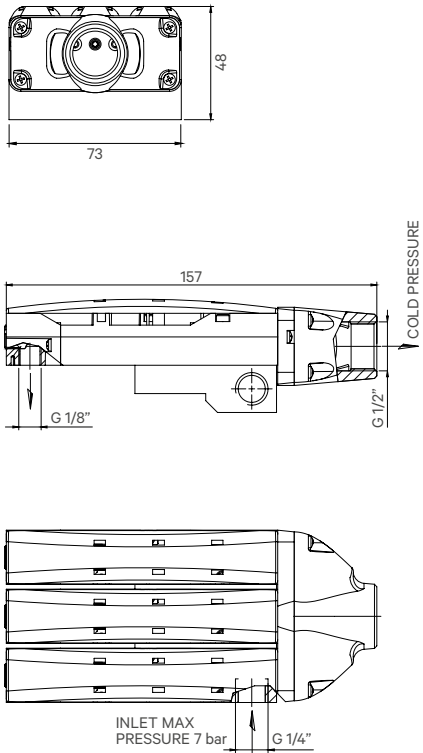
Materials	Body and cover: Nylon 6.6
	Air connections and nozzles: brass
Air inlet port	G-1/4" F
Outlet port (cold flow)	3 x G-1/8" F
Exhaust port (hot flow)	3 x G-1/8" F
Recommended hose	Ø-8x1
Air supply pressure	1 ÷ 7 bar
Cooling power*	360 W - 300 Kcal/h - 1200 BTUH
Optional magnetic support	KACM-VR300

*with inlet pressure 7 Bar and inlet temperature 20°C

PERFORMANCES AND CONSUMPTION TABLE (with air temperature at inlet 20°C)

Pressure bar	Outlet temperature cold flow °C	Consumption NL/min
1	-1,5	96
2	-8	159
3	-15	222
4	-21,5	282
5	-24,5	345
6	-26,5	405
7	-28	462

SERIES VR-300U • SINGLE OUTLET
MODULAR PNEUMATIC COOLERS



GENERAL FEATURES - VR-300U

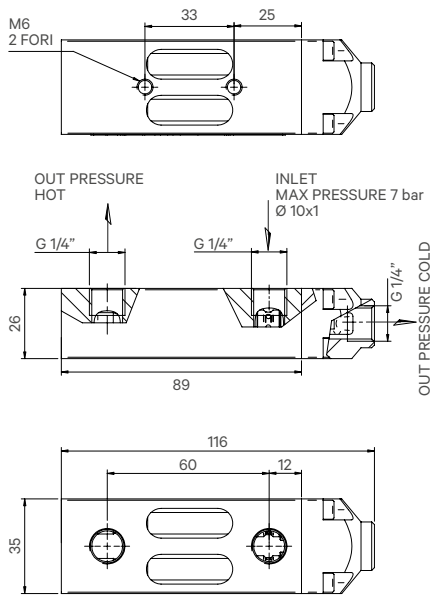
Materials	Body and cover: Nylon 6.6
	Air connections and nozzles: brass
Air inlet port	G-1/4" F
Outlet port (cold flow)	1 x G-1/2" F
Exhaust port (hot flow)	3 x G-1/8" F
Recommended hose	Ø-8x1
Air supply pressure	3 ÷ 7 bar
Cooling power*	360 W - 300 Kcal/h - 1200 BTUH
Optional magnetic support	KACM-VR300

*with inlet pressure 7 Bar and inlet temperature 20°C

PERFORMANCES AND CONSUMPTION TABLE (with air temperature at inlet 20°C)

Pressure bar	Outlet temperature cold flow °C	Consumption NL/min
1	-1,5	96
2	-8	159
3	-15	222
4	-21,5	282
5	-24,5	345
6	-26,5	405
7	-28	462

SERIES VR-200U • SINGLE OUTLET
HIGH-PERFORMANCE COMPACT PNEUMATIC COOLERS



GENERAL FEATURES - VR-200U

Materials	Body and cover: Delrin Ports and nozzles: Brass
Air inlet port	G-1/4" F
Outlet port (cold flow)	G-1/4" F
Exhaust port (hot flow)	G-1/4" F
Recommended hose	Ø-8x1
Air supply pressure	max 7 bar
Cooling power*	264 W - 220 Kcal/h - 880 BTUH
Optional magnetic support	By means of 2 threads M6 on the body
Weight	210 g

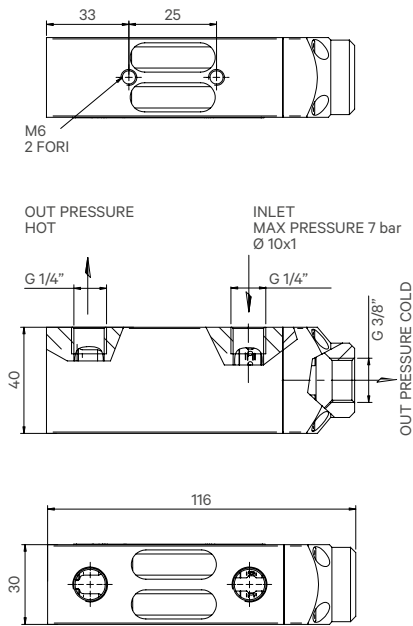
* With inlet pressure 7 Bar and inlet temperature 20°C.

PERFORMANCES AND CONSUMPTION TABLE (with air temperature at inlet 20°C)

Pressure bar	Outlet temperature cold flow °C	Consumption NL/min	Noise level* dBA
1	-2	64	50
2	-12	106	54
3	-18	148	58
4	-23	188	61
5	-26	230	65
6	-28	270	68
7	-31	308	70

*Test made with insulated LOC-LINE flexible tube at cold outlet, and tube L= 1 m at hot air outlet.

SERIES VR-400U • SINGLE OUTLET
HIGH-PERFORMANCE COMPACT PNEUMATIC COOLERS



GENERAL FEATURES - VR-400U

Materials	Body and cover: Delrin Ports and nozzles: Brass
Air inlet port	G-1/4" F
Exhaust port (hot flow)	G-1/4" F
Recommended hose	Ø-10x1
Supply pressure	max 7 bar
Cooling power*	528 W - 440 Kcal/h - 1760 BTUH
Optional fixation of magnetic kit	By means of 2 threads M6 on body
Weight	285 g

* With inlet pressure 7 Bar and inlet temperature 20°C.

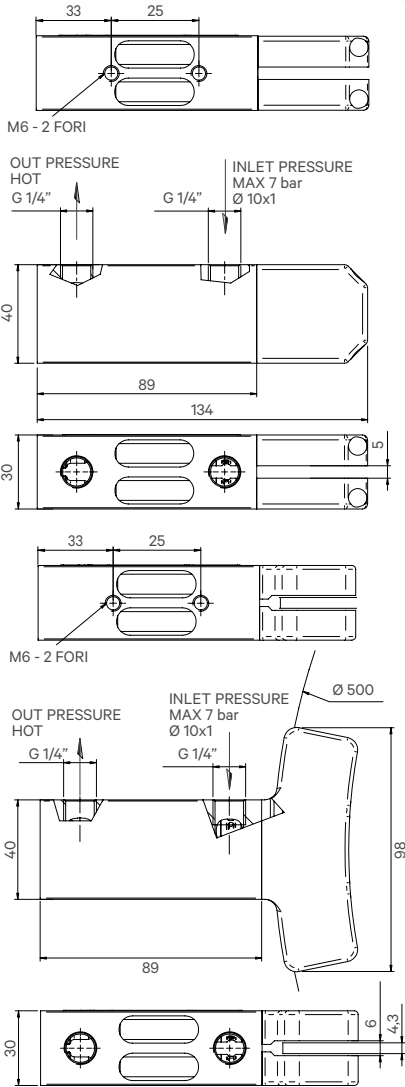
PERFORMANCES AND CONSUMPTION TABLE (with air temperature at inlet 20°C)

Pressure bar	Outlet temperature cold flow °C	Consumption NL/min	Noise level* dBA
1	-2	128	55
2	-12	212	60
3	-18	296	64
4	-23	376	67
5	-26	460	70
6	-28	540	71
7	-31	616	73

*Test made with insulated LOC-LINE flexible tube at cold outlet, and tube L= 1 m at hot air outlet.

SERIES VR-400G • TO COOL DOWN BLADES, BELTS, AND BANDS

HIGH-PERFORMANCE COMPACT PNEUMATIC COOLERS



GENERAL FEATURES - VR-400G

Materials	Body: Derlin Clamps: ABS (other materials on request) Inner spindles: brass
Air inlet port	G-1/4" F
Clamps width (cold flow)	5 mm (customised dimensions on request)
Outlet port (hot flow)	G-1/4" F
Recommended hose	Ø-10x1
Air supply pressure	max 7 bar
Cooling power*	528 W - 440 Kcal/h - 1760 BTUH
Fixation	By means of two M6 threads on body
Weight	340 g

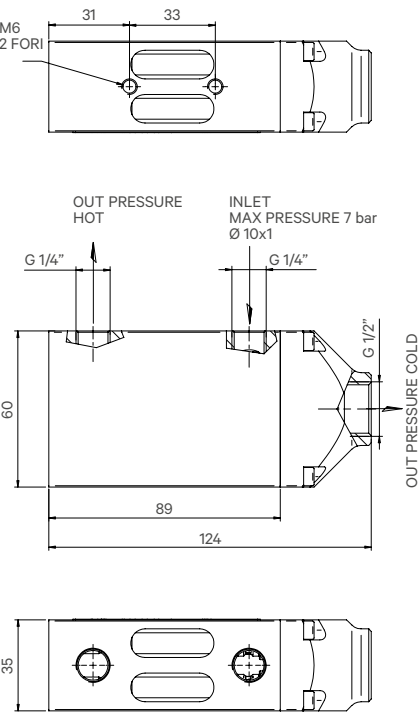
* With inlet pressure 7 Bar and inlet temperature 20°C..

PERFORMANCES AND CONSUMPTION TABLE (with air temperature at inlet 20°C)

Pressure bar	Outlet temperature cold flow °C	Consumption NL/min
1	-2	128
2	-12	212
3	-18	296
4	-23	376
5	-26	460
6	-28	540
7	-31	616

SERIES VR-600U • SINGLE OUTLET

HIGH-PERFORMANCE COMPACT PNEUMATIC COOLERS



GENERAL FEATURES - VR-600U

Materials	Body: Derlin Ports and nozzles: Brass
Air inlet port	G-1/4" F
Clamps width (cold flow)	G-1/2" F
Outlet port (hot flow)	G-1/4" F
Recommended hose	Ø-10x1
Air supply pressure	max 7 bar
Cooling power*	720 W - 600 Kcal/h - 2400 BTUH
Fixation	By means of two M6 threads on body
Weight	460 g

* With inlet pressure 7 Bar and inlet temperature 20°C..

PERFORMANCES AND CONSUMPTION TABLE (with air temperature at inlet 20°C)

Pressure bar	Outlet temperature cold flow °C	Consumption NL/min	Noise level* dBA
1	-2	192	58
2	-12	318	64
3	-18	444	68
4	-23	564	72
5	-26	690	75
6	-28	810	78
7	-31	924	80

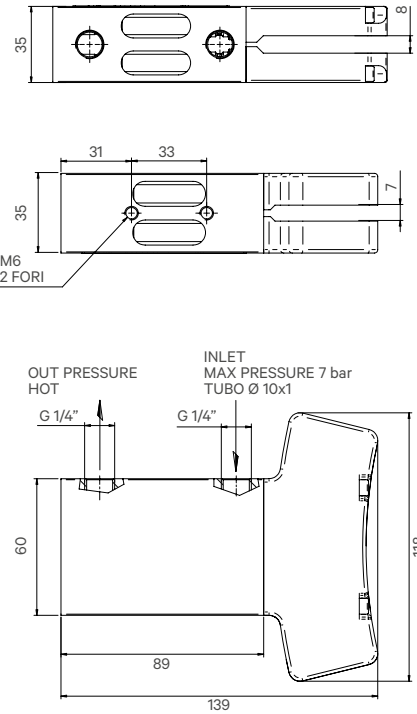
*Test made with insulated LOC-LINE flexible tube at cold outlet, and tube L= 1 m at hot air outlet.

SERIES VR-600G • TO COOL DOWN BLADES, BELTS, AND BANDS

HIGH-PERFORMANCE COMPACT PNEUMATIC COOLERS



SPECIAL VERSION FEATURING INLET FOR MINIMAL LUBRICATION



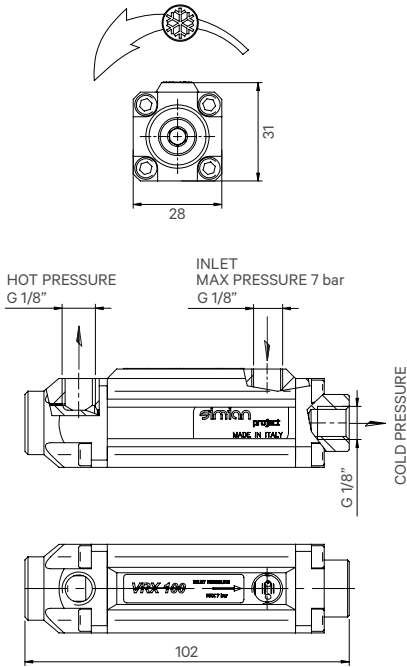
GENERAL FEATURES- VR-600G	
Materials	Body: Derlin Clamps: ABS (other materials on request) Inner spindles: Brass
Air inlet port	G-1/4" F
Clamps width (cold flow)	11 mm (customised dimensions on request)
Outlet port (hot flow)	G-1/4" F
Recommended hose	Ø-10x1
Air supply pressure	max 7 bar
Cooling power*	790 W - 660 Kcal/h - 2640 BTUH
Fixation	By means of two M6 threads on body
Weight	540 g

* With inlet pressure 7 Bar and inlet temperature 20°C..

PERFORMANCES AND CONSUMPTION TABLE (with air temperature at inlet 20°C)		
Pressure bar	Outlet temperature cold flow °C	Consumption NL/min
1	-2	192
2	-12	318
3	-18	444
4	-23	564
5	-26	690
6	-28	810
7	-31	924

SERIES VRX-100

HIGH-PERFORMANCE PNEUMATIC COOLERS



GENERAL FEATURES - VRX-100	
Materials	Sleeve: anodized aluminium Ends: Nylon 6.6
Air inlet port	G-1/8" F
Outlet port (cold flow)	G-1/8" F
Exhaust port (hot flow)	G-1/8" F
Recommended hose	Ø-8x1
Air supply pressure	1 ÷ 7 bar
Cooling power*	132 W - 110 Kcal/h - 440 BTUH
Optional magnetic support	KACM-VRX-100
Weight	170 g

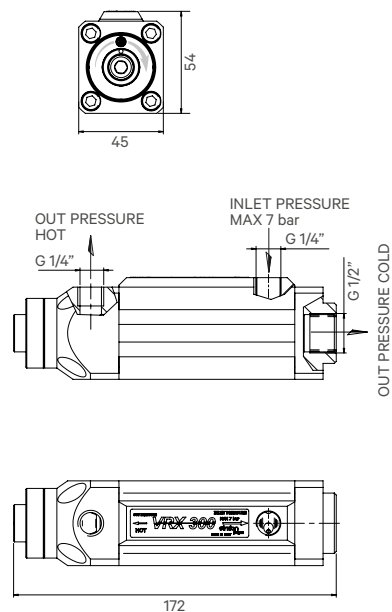
* With inlet pressure 7 Bar and inlet temperature 20°C..

PERFORMANCES AND CONSUMPTION TABLE (with air temperature at inlet 20°C)			
Pressure bar	Outlet temperature cold flow °C	Consumption NL/min	Noise level* dBA
1	-2	32	54
2	-12	53	58
3	-18	74	62
4	-23	94	64
5	-26	115	64
6	-28	135	66
7	-31	154	68

*Test made with insulated LOC-LINE flexible tube at cold outlet, and tube L= 1 m at hot air outlet.

SERIES VRX-300

HIGH-PERFORMANCE PNEUMATIC COOLERS



GENERAL FEATURES - VRX-300	
Materials	Sleeve: anodized aluminium Ends: Delrin100
Air inlet port	G-1/4" F
Outlet port (cold flow)	G-1/2" F
Exhaust port (hot flow)	G-1/4" F
Recommended hose	Ø 10x1
Air supply pressure	1 ÷ 7 bar
Cooling power*	600 W - 523 Kcal/h - 2075 BTUH
Optional magnetic support	KACM-VRX500
Weight	740 g

* With inlet pressure 7 Bar and inlet temperature 20°C..

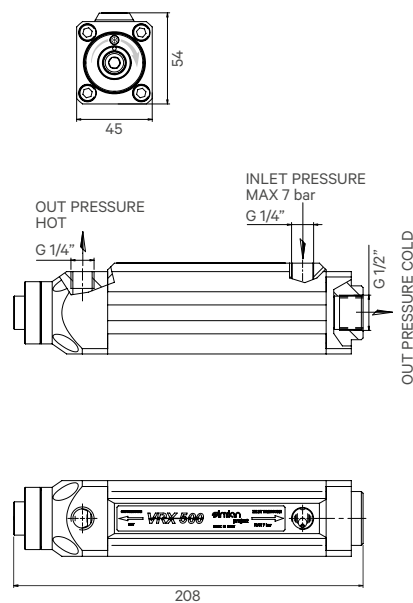
PERFORMANCES AND CONSUMPTION TABLE (with air temperature 20°C)

Pressure bar	Outlet temperature cold flow °C	Consumption NL/min	Noise level* dBA
1	-3	50	67
2	-7	170	72
3	-10	290	74
4	-13	410	76
5	-16	525	78
6	-17	650	80
7	-19	750	82

*Test made with insulated LOC-LINE flexible tube at cold outlet, and tube L= 1 m at hot air outlet.

SERIES VRX-500

HIGH-PERFORMANCE PNEUMATIC COOLERS



GENERAL FEATURES - VRX-500	
Materials	Sleeve: anodized aluminium Ends: Delrin100
Air inlet port	G-1/4" F
Outlet port (cold flow)	G-1/2" F
Exhaust port (hot flow)	G-1/4" F
Recommended hose	Ø 10x1
Air supply pressure	1 ÷ 7 bar
Cooling power*	730 W - 630 Kcal/h - 2500 BTUH
Optional magnetic support	KACM-VRX500
Weight	860 g

* With inlet pressure 7 Bar and inlet temperature 20°C..

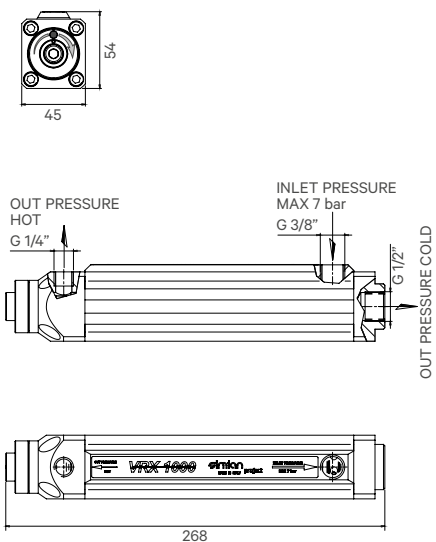
PERFORMANCES AND CONSUMPTION TABLE (with air temperature 20°C)

Pressure bar	Outlet temperature cold flow °C	Consumption NL/min	Noise level* dBA
1	-3	120	66
2	-7	250	71
3	-10	380	72
4	-13	500	75
5	-16	633	77
6	-17	783	78
7	-19	900	80

*Test made with insulated LOC-LINE flexible tube at cold outlet, and tube L= 1 m at hot air outlet.

SERIES VRX-1000
HIGH-PERFORMANCE PNEUMATIC COOLERS

ACCESSORIES
PNEUMATIC COOLERS



GENERAL FEATURES - VRX-1000	
Materials	Sleeve: anodized aluminium Ends: Delrin100
Air inlet port	G-3/8" F
Outlet port (cold flow)	G-1/2" F
Exhaust port (hot flow)	G-1/4" F
Recommended hose	Ø 12x1
Air supply pressure	1 ÷ 7 bar
Cooling power*	1650 W - 1417 Kcal/h - 5600 BTUH
Optional magnetic support	KACM-VRX1000
Weight	1060 g

* With inlet pressure 7 Bar and inlet temperature 20°C..

PERFORMANCES AND CONSUMPTION TABLE (with air temperature at inlet 20°C)

Pressure bar	Outlet temperature cold flow °C	Consumption NL/min	Noise level* dBA
1	-3	230	70
2	-7	500	73
3	-10	800	75
4	-13	1100	77
5	-16	1424	79
6	-17	1760	81
7	-19	2025	83

*Test made with insulated LOC-LINE flexible tube at cold outlet, and tube L= 1 m at hot air outlet.



MAGNETIC SUPPORT	
Part-number	Cooler
KACM-VR100	VR100
KACM-VR200	VR200
KACM-VR300	VR300
KACM-VR246	VR200U-G / VR400U-G / VR600U-G
KACM-VRX100	VRX100
KACM-VRX300 / VRX500	VRX300 / VRX500
KACM-VRX1000	VRX1000

ADJUSTABLE NOZZLE (INSULATED VERSION) FOR COLD OUTLET				
Part-number	Port Ø	Nozzle Ø	No. modules	Length mm
AC28	1/8"	3	4	100
AC34	1/4"	3	4	100
AC47	3/8"	6	6	180
AC27	1/2"	6	6	180

ADJUSTABLE NOZZLE (NON-INSULATED VERSION) FOR COLD OUTLET				
Part-number	Port Ø	Nozzle Ø	No. modules	Length mm
82021/8 1/8-3	1/8"	3	8	155
84041/6 1/2-9	1/2"	9	6	170

Other configurations available on request

STRAIGHT PUSH-IN FITTING FOR AIR SUPPLY			
Part-number	Hose Ø	Size	Cooler
S6510	8	1/8"	VR-100 / VR-200 / VR-300 / VRX-100
S6510	10	1/4"	VRX-300 / VRX-500
S6510	12	1/4"	VRX-1000
S6510	10	3/8"	VR-600

ELBOW PUSH-IN FITTING FOR AIR SUPPLY			
Part-number	Hose Ø	Size	Cooler
S6520	8	1/8"	VR-100 / VR-200 / VR-300 / VRX-100
S6520	10	1/4"	VRX-300 / VRX-500
S6520	12	1/4"	VRX-1000
S6520	10	3/8"	VR-600

SILENCER FOR HOT FLOW EXHAUST		
Part-number	Size	Noise at 6 bar dBA
SC 1/8	1/8"	70
SC 1/4	1/4"	67
SC 3/8	3/8"	67

In sintered bronze.

SILENCER FOR COLD FLOW OUTLET (ITEM FOR VRX-300/VRX-500)	
Codice	Size
AC25	1/2"

CONTROL UNITS XTRONIC2

1 / 2 / 3 SOLENOID VALVES

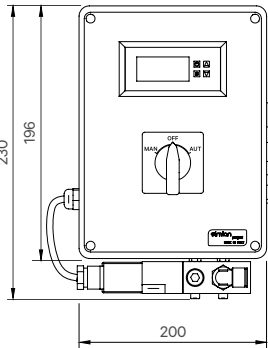
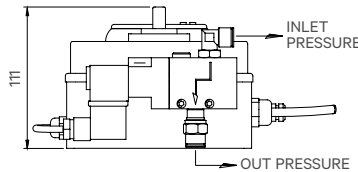
CONTROL UNIT XTRONIC2 C EV-1S/C



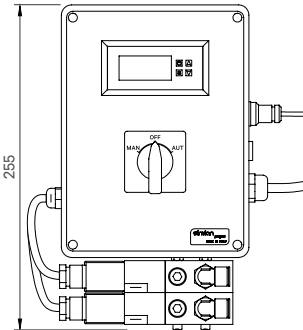
DESCRIPTION

Control box featuring:

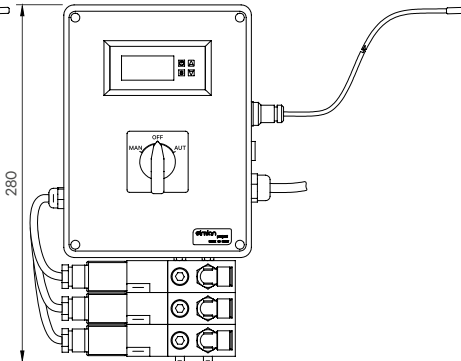
1. Switch for automatic/manual operation
2. Solenoid valve
3. Electronic kit with 7-segment display
4. Temperature probe and safety fuses (high-temperature kit available on request).



XTRONIC2 C-EV-1S/C
1 SOLENOID VALVE



XTRONIC2 C-2EV-1S/C
2 SOLENOID VALVES



XTRONIC2 C-3EV-1S/C
3 SOLENOID VALVES

GENERAL FEATURES - XTRONIC2 (1/2/3 SOLENOID VALVES)

Supply voltage	220 V AC
Recommended hose	Ø 10x1
Supply pressure	max 7 bar
Probe length	1 m
Application	Outside of cabinets
Weight	
XTRONIC2 C-EV-1S/C (1 solenoid valve)	1300 g
XTRONIC2 C-2EV-1S/C (2 solenoid valves)	1600 g
XTRONIC2 C-3EV-1S/C (3 solenoid valves)	1900 g

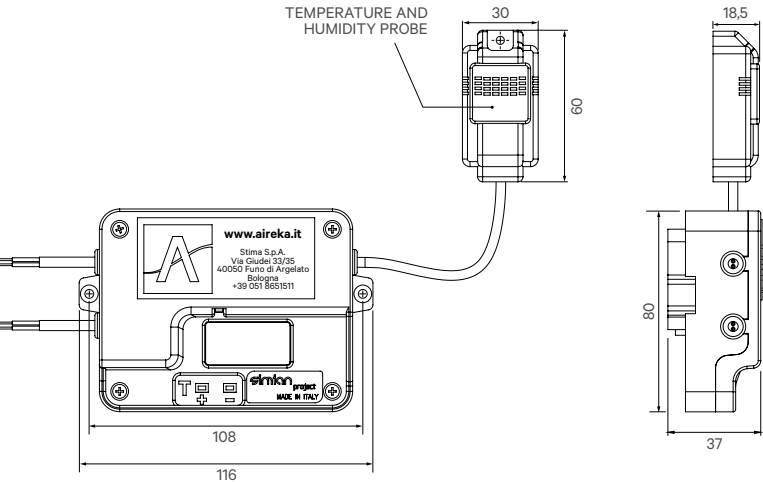
CONTROL UNITS XTRONIC 345 B

PNEUMATIC COOLERS



GENERAL FEATURES - XTRONIC 345 B

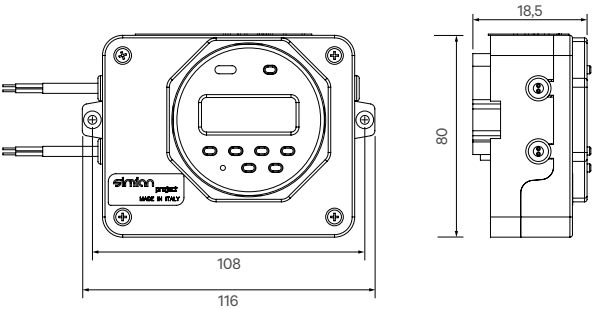
Supply voltage	24 V DC
Probe length	1 m
Temperature range	-20°C +60°C
Humidity range	00% 100% RH
Accuracy	0.1°C , 0.1 % RH
Current capacity	max 10 A
Coil voltage and power	24 V - 3.1 W
Electric wires' section	0.75 mm
Supply cable length	1.5 m
Application	Inside cabinets





GENERAL FEATURES - XTRONIC 345 T

Supply	24 V DC
Probe length	1 m
Time setting	Weekly / h24
Humidity range	0% - 100% RH
Accuracy	0.1°C , 0.1 % RH
Current capacity	max 10 A
Coil power	24 V - 3.1 W
Electric wires' section	0.75 mm
Application	Inside / outside



The installation of coolers in control cabinets can be of 3 types:

- 1. **EXTERNAL unit**
to have the inner temperature of the cabinet displayed directly;
- 2. **INTERNAL unit**
so that to have a compact solution;
- 3. **HYBRID systems** where the pneumatic cooler works in combination with an already existing air conditioning unit.

- Hot air
- Cold air

